

HW Class 6 Q6 (R Functions)

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Q6. How would you generalize the original code above to work with any set of input protein structures?

The code could be generalized by writing a function that removes the repeated steps for each protein. This can be done by taking a PDB ID as an input. This function allows the PDB file to be read, trims the structure to the selected chain and atoms, extracts the B-factor values, and generates a plot for the specified proteins. Additionally, returning the B-factor vector allows the results to be used for more comparison or clustering.

Write your own function starting from the code above that analyzes protein drug interactions by reading in any protein PDB data and outputs a plot for the specified protein.

The code given is generalized below:

```
library(bio3d)
```

```

# analyze_protein()
# Input:
# pdb_id is a character string that represents PDB ID,
# chain_id is the protein chain used to analyze (default= "A"),
# atom_type is the atom type to select (default= "CA")
# Function:
# Reads the protein structure from the PDB database,
# trims to a specified chain and atom type,
# Then it extracts the B-factor values and plots the B-factors
# Output: Numeric vector with B-factor values for the selected atoms
analyze_protein <- function(pdb_id, chain_id = "A", atom_type = "CA") {
  pdb <- read.pdb(pdb_id)
  pdb_trim <- trim.pdb(pdb, chain = chain_id, elety = atom_type)
  b_factors <- pdb_trim$atom$b

  plotb3(b_factors, sse= pdb_trim, typ = "l", ylab = "Bfactor")

  return(b_factors)
}

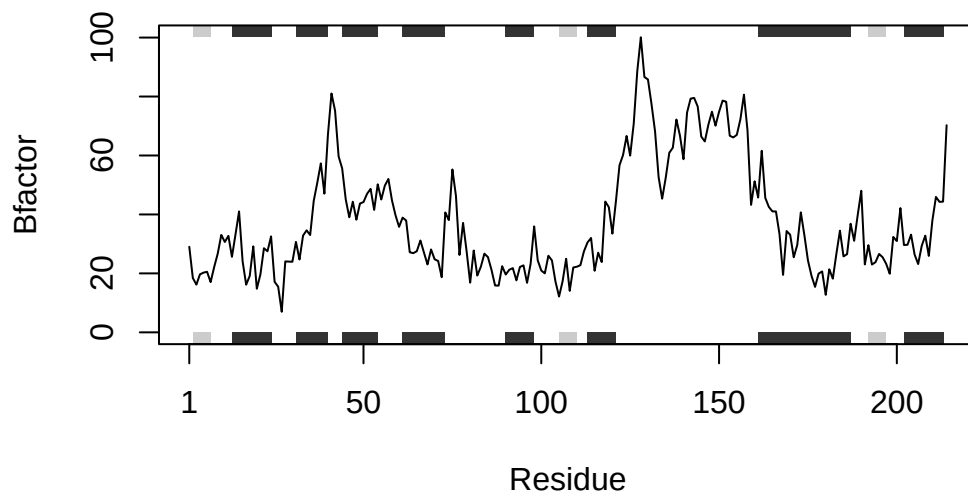
```

```

# Example usage
s1.b <- analyze_protein("4AKE") # kinase with drug

```

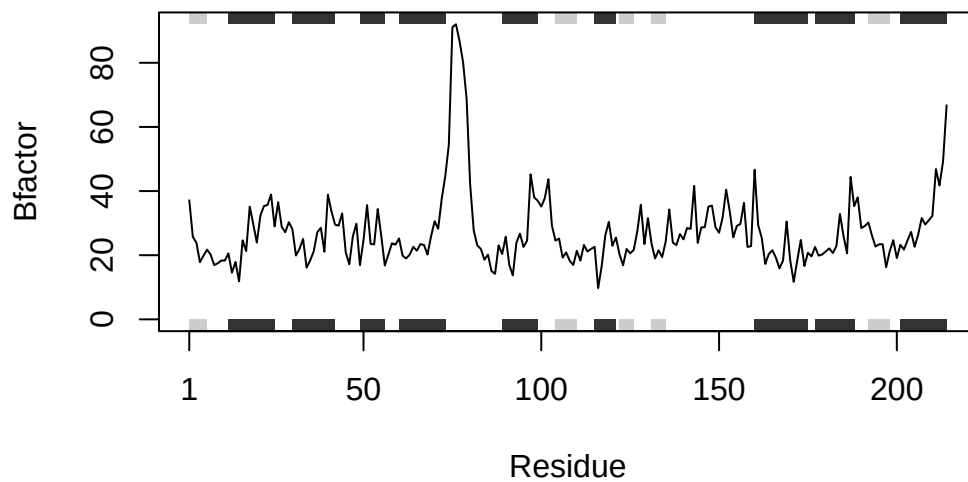
Note: Accessing on-line PDB file



```
s2.b <- analyze_protein("1AKE") # kinase no drug
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE



```
s3.b <- analyze_protein("1E4Y") # kinase with drug
```

Note: Accessing on-line PDB file

